INFORMATION THEORY & CODING

Dr. Bo Ning

Associate Professor, College of Computer Science

Email: bo.ning@nankai.edu.cn

Review Summary

- **Definition** (*Entropy*) The *entropy* $H(X)$ of a discrete random variable X is defined by

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x).$$

- **Properties of H**

1. $H(X) \geq 0$.
2. $H_b(X) = (\log_b a) H_a(X)$.
3. (*Conditioning reduces entropy*) For any two random variables, X and Y, we have $H(X|Y) \leq H(X)$ with equality iff X and Y are independent.
4. (*Independence bound on entropy*) $H(X_1, X_2, \ldots, X_n) \leq \sum_{i=1}^{n} H(X_i)$, with equality iff the X_i's are independent.
5. $H(X) \leq \log |\mathcal{X}|$, with equality iff X is distributed uniformly over $\mathcal{X}$.

Review Summary

- **Definition** (*Relative entropy*) The *relative entropy* $D(p||q)$ of the probability mass function p with respect to the probability mass function q is defined by

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

- **Definition** (*Mutual information*) The *mutual information* between two random variables X and Y is defined as

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

Review Summary

- **Properties of D and I**

1. $I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X,Y)$.
2. $D(p||q) \geq 0$ with equality iff $p(x) = q(x)$, for all $x \in \mathcal{X}$ (*information inequality*).
3. $I(X;Y) = D(p(x,y)||p(x)p(y)) \geq 0$, with equality iff $p(x,y) = p(x)p(y)$ (i.e., X and Y are independent).
4. If $|\mathcal{X}| = m$, and u is the uniform distribution over $\mathcal{X}$, then $D(p||u) = \log m - H(p)$.

- **Jensen's inequality**

If f is a convex function, then $Ef(X) \geq f(EX)$.

Data-processing inequality

If $X \to Y \to Z$ forms a Markov chain, then $I(X;Y) \geq I(X;Z)$.

Fano's inequality

- **Problem 2.5** (*Zero conditional entropy*) Show that if $H(X|Y)=0$, then X is a function of Y, i.e., for all y with $p(y)>0$, there is only one possible value of x with $p(x,y)>0$.

  Proof. Assume that there exists an y, say y_0 and two different values of x, say x_1 and x_2 such that $p(y_0,x_1)>0$ and $p(y_0,x_2)>0$. Then $p(y_0)\geq p(y_0,x_1)+p(y_0,x_2)>0$, and $p(x_1|y_0)$ and $p(x_2|y_0)$ are not equal to 0 or 1. Thus,

$$
\begin{aligned}
H(X|Y) &= -\sum_y p(y)\sum_x p(x|y)\log p(x|y)\\
&\geq p(y_0)(-p(x_1|y_0)\log p(x_1|y_0)-p(x_2|y_0)\log p(x_2|y_0))\\
&> 0,
\end{aligned}
$$

  since $-t\log t\geq 0$ for $0\leq t\leq 1$, and is strictly positive for $t\neq 0,1$, which is a contradiction to $H(X|Y)=0$.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y. Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y. Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.
- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y. Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.
- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

Why do we need to related P_e to the entropy $H(X|Y)$?

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y. Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.
- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

Why do we need to related P_e to the entropy $H(X|Y)$?

When we have a communication system, we send X, but receive a corrupted version Y. We want to infer X from Y. Our estimate is $\hat{X}$ and we will make a mistake as

$$P_e = \Pr[\hat{X} \neq X]$$

Markov chain $X \to Y \to \hat{X}$.

Fano's inequality

- **Problem.** A random variable Y is related to another random variable X with a distribution $p(x)$. From Y, we calculate a function $g(Y) = \hat{X}$, where $\hat{X}$ is an estimate of X and takes on values in $\hat{\mathcal{X}}$. We observe that $X \to Y \to \hat{X}$ forms a Markov chain. How to bound the estimate error probability $P_e = \Pr[\hat{X} \neq X]$?

Fano's inequality

- **Theorem 2.10.1** (*Fano's inequality*) For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

This inequality can be weakened to

$$1 + P_e \log(|\mathcal{X}| - 1) \geq H(X|Y)$$

or

$$P_e \geq \frac{H(X|Y) - 1}{\log|\mathcal{X}| - 1}.$$

Proof.

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

Proof. Define an error random variable as

$$E = \begin{cases} 1 & \text{if } \hat{X} \neq X, \\ 0 & \text{if } \hat{X} = X. \end{cases}$$

Using the chain rule for entropies to expand $H(E, X|\hat{X})$ in two different ways, we have

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}|-1)}. \end{aligned}$$

Since *conditioning reduces entropy*, $H(E|\hat{X}) \leq H(E) = H(P_e)$. Since E is a function of X and $\hat{X}$, the conditional entropy $H(E|X, \hat{X})$ is equal to 0. We now look at $H(X|E, \hat{X})$. By the equation $H(X|Y) = \sum_y p(y) H(X|Y = y)$, we have

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{\Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1)\}. \end{aligned}$$

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

Proof.

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}|-1)} . \end{aligned}$$

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{\Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1)\}. \end{aligned}$$

By definition of E, X is conditionally deterministic given $\hat{X} = \hat{x}$ and $E = 0$, then $H(X|\hat{X} = \hat{x}, E = 0) = 0$. If $\hat{X} = \hat{x}$ and $E = 1$, then X must take a value in the set $\{x \in \mathcal{X} : x \neq \hat{x}\}$ which contains $|\mathcal{X}| - 1$ elements. Then $H(X|\hat{X} = \hat{x}, E = 1) \leq \log(|\mathcal{X}| - 1)$.

$$\begin{aligned} H(X|E, \hat{X}) &\leq \sum_{\hat{x} \in \mathcal{X}} \Pr[\hat{X} = \hat{x}, E = 1] \log(|\mathcal{X}| - 1) \\ &= \Pr[E = 1] \log(|\mathcal{X}| - 1) \\ &= P_e \log(|\mathcal{X}| - 1). \end{aligned}$$

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

Proof.

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}|-1)}. \end{aligned}$$

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{\Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1)\}. \end{aligned}$$

$$\begin{aligned} H(X|E, \hat{X}) &\leq \sum_{\hat{x} \in \mathcal{X}} \Pr[\hat{X} = \hat{x}, E = 1] \log(|\mathcal{X}| - 1) \\ &= \Pr[E = 1] \log(|\mathcal{X}| - 1) \\ &= P_e \log(|\mathcal{X}| - 1). \end{aligned}$$

By the data-processing inequality, we have $I(X; \hat{X}) \leq I(X; Y)$ and therefore $H(X|\hat{X}) \geq H(X|Y)$.

Corollary

- **Corollary** For any two random variables X and Y, let $p = \Pr(X \neq Y)$.

$$H(p) + p \log(|\mathcal{X}| - 1) \geq H(X|Y).$$

Proof. Let $\hat{X} = Y$ in Fano's inequality.

Fano's inequality

- **Remark** Suppose that there is no knowledge of Y. Thus, X must be guessed without any information. Let $X \in \{1, 2, \ldots, m\}$ and $p_1 \geq p_2 \geq \cdots \geq p_m$. Then the best guess of X is $\hat{X} = 1$ and the resulting probability of error is $P_e = 1 - p_1$. Fano's inquality becomes

$$H(P_e) + P_e \log(m-1) \geq H(X).$$

The probability mass function

$$(p_1, p_2, \cdots, p_m) = (1 - P_e, \frac{P_e}{m-1}, \cdots, \frac{P_e}{m-1})$$

achieves this bound with equality.

Fano's inequality

由此得到如下的不等式：

引理 2.10.1　如果 X 和 X' 独立同分布，具有熵 $H(X)$，则

$$\Pr(X=X')\geqslant 2^{-H(X)}$$

当且仅当 X 服从均匀分布，等号成立。

证明：假定 $X\sim p(x)$。由 Jensen 不等式，可得

$$2^{E\log p(X)}\leqslant E2^{\log p(X)}$$

含义是

$$2^{-H(X)}=2^{\sum p(x)\log p(x)}\leqslant\sum p(x)2^{\log p(x)}=\sum p^2(x)$$

推论　设 X 和 X' 相互独立，且 $X\sim p(x)$，$X'\sim r(x)$，$x,x'\in\mathcal{X}$，那么

$$\Pr(X=X')\geqslant 2^{-H(p)-D(p\|r)}$$

$$\Pr(X=X')\geqslant 2^{-H(r)-D(r\|p)}$$

证明：我们有

$$\begin{aligned}2^{-H(p)-D(p\|r)}&=2^{\sum p(x)\log p(x)+\sum p(x)\log\frac{r(x)}{p(x)}}\\&=2^{\sum p(x)\log r(x)}\\&\leqslant\sum p(x)2^{\log r(x)}\\&=\sum p(x)r(x)\\&=\Pr(X=X')\end{aligned}$$

其中的不等式可由 Jensen 不等式和函数 $f(y)=2^y$ 得到。

Fano's inequality

- **Lemma 2.10.1** If X and X' are i.i.d. with entropy $H(X)$,

$$\Pr[X = X'] \geq 2^{-H(X)},$$

with equality iff X has a unifrom distribution.

- **Corollary** Let X, X' be independent with $X \sim p(x)$, $X' \sim r(x)$, $x, x' \in \mathcal{X}$. Then

$$\Pr[X = X'] \geq 2^{-H(p)-D(p||r)}$$
$$\Pr[X = X'] \geq 2^{-H(r)-D(r||p)}$$

Stock Market

- Initial investment Y_0, daily return ratio r_i, in the t-th day, your money is
$$Y_t = Y_0 r_1 \cdot \ldots \cdot r_t.$$

- Now if returns ratio r_i's are i.i.d., with
$$r_i = \begin{cases} 4, & w.p.\ 1/2 \\ 0, & w.p.\ 1/2 \end{cases}$$

- So you think the expected return ratio is $Er_i = 2$.

- and then
$$EY_t = E(Y_0 r_1 \cdot \ldots \cdot r_t) = Y_0 (Er_i)^t = Y_0 2^t ???$$

Stock Market

- With $Y_0 = 1$, actual return Y_t goes like

$$1 \quad 4 \quad 16 \quad 0 \quad 0 \quad \cdots$$

- Why?: products does not behave the same as addition!

The Markov and Chebyshev Inequalities

- **Theorem** (*Markov Inequality*) Suppose that X is a random variable such that $\Pr[X \geq 0] = 1$. Then for every real number $t > 0$,

$$\Pr[X \geq t] \leq \frac{EX}{t}.$$

Proof.

$$EX = \sum_{x} xp(x) = \sum_{x<t} xp(x) + \sum_{x \geq t} xp(x).$$

Since X can have only nonnegative values, we have

$$EX \geq \sum_{x \geq t} xp(x) \geq \sum_{x \geq t} tp(x) = t\Pr[X \geq t].$$

The Markov and Chebyshev Inequalities

- **Theorem** (*Chebyshev Inequality*) Let X be a random variable for which $\mathrm{Var}(X)$ exists. Then for every number $t > 0$,

$$\Pr[|X - EX| \geq t] \leq \frac{\mathrm{Var}(X)}{t^2}.$$

Proof. Let $Y = (X - EX)^2$. Then $\Pr[Y \geq 0] = 1$ and $EY = \mathrm{Var}(X)$. By applying the Markov inequality to Y, we obtain:

$$\Pr[|X - EX| \geq t] = \Pr[Y \geq t^2] \leq \frac{\mathrm{Var}(X)}{t^2}.$$

Weak Law of Large Numbers

- **Theorem** (*Weak Law of Large Numbers*) Suppose that $X_1, X_2, \ldots, X_n$ are n independent, identically distributed (i.i.d.) random variables, then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \rightarrow EX \qquad \text{in probability},$$

i.e., for every number $\epsilon > 0$,

$$\lim_{n\rightarrow\infty} \Pr[\big|\frac{1}{n}\sum_{i=1}^{n} X_i - EX\big| \leq \epsilon] = 1.$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^{n} X_i}$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^{n} X_i}$$

- We can write

$$X_i = e^{\log X_i}$$

Hence

$$\sqrt[n]{\prod_{i=1}^{n} X_i} = e^{\frac{1}{n}\sum_{i=1}^{n} \log X_i}$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^{n} X_i}$$

- We can write

$$X_i = e^{\log X_i}$$

Hence

$$\sqrt[n]{\prod_{i=1}^{n} X_i} = e^{\frac{1}{n}\sum_{i=1}^{n} \log X_i}$$

$$\sqrt[n]{\prod_{i=1}^{n} X_i} \to e^{E(\log X)} \leq e^{\log EX} = EX.$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^{n} X_i}$$

- We can write

$$X_i = e^{\log X_i}$$

Hence

$$\sqrt[n]{\prod_{i=1}^{n} X_i} = e^{\frac{1}{n}\sum_{i=1}^{n} \log X_i}$$

$$\sqrt[n]{\prod_{i=1}^{n} X_i} \to e^{E(\log X)} \le e^{\log EX} = EX.$$

geometric mean $\sqrt[n]{\prod_{i=1}^{n} X_i} \le$ *arithmetic mean* $\frac{1}{n}\sum_{i=1}^{n} X_i$

L.L.N for Product of Random Variables

- Stock example:

$$E \log r_i = \frac{1}{2} \log 4 + \frac{1}{2} \log 0 = -\infty$$

$$E(Y_t) \to Y_0 e^{E \log r_i} = 0, \quad t \to \infty$$

L.L.N for Product of Random Variables

- Stock example:

$$E \log r_i = \frac{1}{2}\log 4 + \frac{1}{2}\log 0 = -\infty$$

$$E(Y_t) \to Y_0 e^{E \log r_i} = 0, \quad t \to \infty$$

- Example:

$$X = \begin{cases} a & \text{w.p. } 1/2 \\ b & \text{w.p. } 1/2 \end{cases}$$

$$E\left(\sqrt[n]{\prod_{i=1}^n X_i}\right) \to \sqrt{ab} \le \tfrac{a+b}{2}$$

Asymptotic Equipartition Property (AEP)

- **Definition** (*Convergence of random variables*). Given a sequence of random variables, $X_1, X_2, \ldots$, we say that the sequence $X_1, X_2, \ldots$ converges to a random variable X:

  1. *In probability* if for every $\epsilon > 0$, $\Pr[|X_n - X| \geq \epsilon] \to 0$
  2. *In mean square* if $E(X_n - X)^2 \to 0$
  3. *With probability 1* (a.k.a. *almost surely*) if $\Pr[\lim_{n\to\infty} X_n = X] = 1$

Asymptotic Equipartition Property (AEP)

- **Theorem 3.1.1** (*AEP*) If $X_1, X_1, \ldots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n}\log p(X_1, X_2, \ldots, X_n) \to H(X) \qquad \text{in probability.}$$

Asymptotic Equipartition Property (AEP)

- **Theorem 3.1.1** (*AEP*) If $X_1, X_1, \ldots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n}\log p(X_1, X_2, \ldots, X_n) \to H(X) \qquad \text{in probability.}$$

Proof. Since X_i's are i.i.d., so are $\log p(X_i)$'s. Hence, by the *weak law of large numbers*,

$$\begin{aligned} -\frac{1}{n}\log p(X_1, X_2, \ldots, X_n) &= -\frac{1}{n}\sum_i \log p(X_i) \\ &\to -E\log p(X) \qquad \text{in probability} \\ &= H(X), \end{aligned}$$

AEP, Typical Set

- AEP lies in the *heart* of information theory.
  - ◇ Proof for lossless source coding
  - ◇ Proof for channel capacity
  - ◇ and more...

AEP, Typical Set

- AEP lies in the *heart* of information theory.
  - Proof for lossless source coding
  - Proof for channel capacity
  - and more...

- **Definition** A *typical set* $A_\epsilon^{(n)}$ contains all sequences $(x_1, x_2, \ldots, x_n) \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \leq p(x_1, x_2, \ldots, x_n) \leq 2^{-n(H(X)-\epsilon)}.$$

Properties of AEP

- **Theorem 3.1.2**

  1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, then
     $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \ldots, x_n) \leq H(X) + \epsilon$.
  2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
  3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A.
  4. $|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Properties of AEP

- **Theorem 3.1.2**

  1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then
     $H(X) - \epsilon \le -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \le H(X) + \epsilon$.
  2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
  3. $|A_\epsilon^{(n)}| \le 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A.
  4. $|A_\epsilon^{(n)}| \ge (1-\epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 1. Immediate from the definition of $A_\epsilon^{(n)}$.

The number of bits used to describe sequences in typical set is approximately $nH(X)$.

Properties of AEP

- **Theorem 3.1.2**

1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, then
   $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \ldots, x_n) \leq H(X) + \epsilon$.
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A.
4. $|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 2. By Theorem 3.1.1, the probability of the event $(X_1, X_2, \ldots, X_n) \in A_\epsilon^{(n)}$ tends to 1 as $n \to \infty$. Thus, for any $\delta > 0$, there exists an n_0 such that for all $n \geq n_0$, we have

$$\Pr\{\left| -\frac{1}{n} \log p(X_1, X_2, \ldots, X_n) - H(X)\right| < \epsilon\} > 1 - \delta.$$

Setting $\delta = \epsilon$, the conclusion follows.

Properties of AEP

- **Theorem 3.1.2**

1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, then
   $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \ldots, x_n) \leq H(X) + \epsilon$.
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A.
4. $|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 3.

$$\begin{aligned}
1 &= \sum_{\mathbf{x} \in \mathcal{X}^n} p(\mathbf{x}) \geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x}) \\
&\geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X)+\epsilon)} \\
&= 2^{-n(H(X)+\epsilon)} |A_\epsilon^{(n)}|.
\end{aligned}$$

Properties of AEP

- **Theorem 3.1.2**

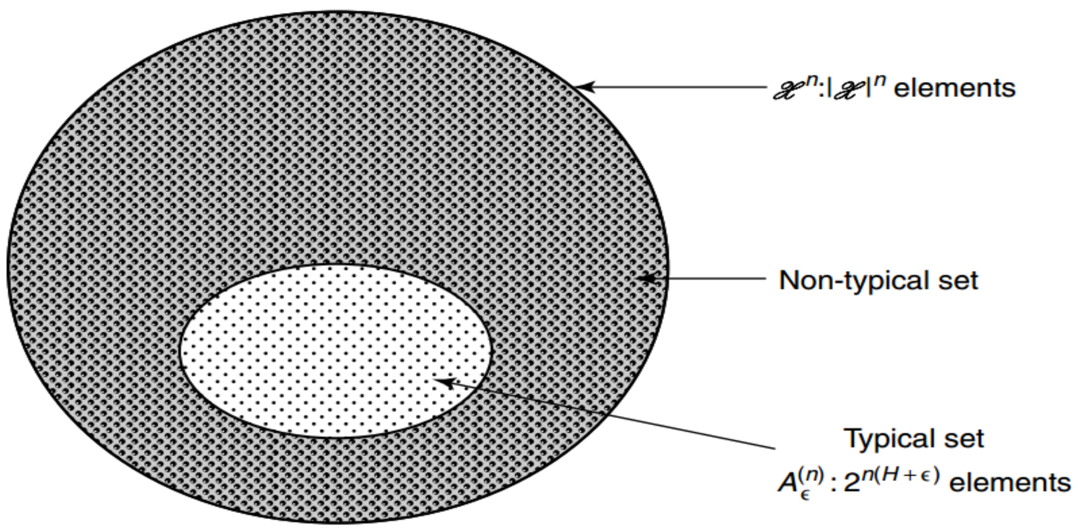
  1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, then
     $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \ldots, x_n) \leq H(X) + \epsilon$.
  2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
  3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A.
  4. $|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 4. For sufficiently large n, $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$, so that

$$\begin{aligned} 1 - \epsilon &< \Pr[A_\epsilon^{(n)}] \\ &\leq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X)-\epsilon)} \\ &= 2^{-n(H(X)-\epsilon)} |A_\epsilon^{(n)}|. \end{aligned}$$

Typical set disgram

- This enables us to divide all sequences into two sets
  - Typical set: high probability to occur, sample entropy is close to true entropy
    so we will focus on analyzing sequences in typical set
  - Non-typical set: small probability, can ignore in general



Asymptotic Equipartition Property (AEP)

- **Theorem 3.2.1** Let X^n be i.i.d. $\sim p(x)$. Let $\epsilon > 0$. Then there exists a code that maps sequences x^n of length n into binary strings such that the mapping is one-to-one (and therefore invertible) and

$$E\left[\frac{1}{n}\ell(X^n)\right] \leq H(X) + \epsilon$$

for n sufficiently large.

Asymptotic Equipartition Property (AEP)

- **Theorem 3.2.1**

$$E\Big[\frac{1}{n}\ell(X^n)\Big] \leq H(X) + \epsilon$$

for n sufficiently large.

Proof. Description in *typical set* requires no more than $n(H(X)+\epsilon)+1$ bits (correction of 1 bit because of integrality).

Description in *atypical set* $A_\epsilon^{(n)^C}$ requires no more than $n\log|\mathcal{X}|+1$ bits.

Add another bit to indicate whether in $A_\epsilon^{(n)}$ or not to get whole description.

- **Theorem 3.2.1**

$$E\Big[\frac{1}{n}\ell(X^n)\Big] \le H(X) + \epsilon$$

for n sufficiently large.

Proof. (cont.) Let $\ell(x^n)$ be the length of the binary description of x^n. Then, $\forall \epsilon > 0$, there exists n_0 s.t. $\forall n > n_0$,

$$\begin{aligned}
E(\ell(X^n)) &= \sum_{x^n} p(x^n)\ell(x^n) \\
&= \sum_{x^n \in A_\epsilon^{(n)}} p(x^n)\ell(x^n) + \sum_{x^n \in A_\epsilon^{(n)C}} p(x^n)\ell(x^n) \\
&\le \sum_{x^n \le A_\epsilon^{(n)}} p(x^n)(n(H+\epsilon)+2) \\
&\quad + \sum_{x^n \in A_\epsilon^{(n)C}} p(x^n)(n\log|\mathcal{X}|+2) \\
&= \Pr[A_\epsilon^{(n)}](n(H+\epsilon)+2) + \Pr[A_\epsilon^{(n)^C}](n\log|\mathcal{X}|+2) \\
&\le n(H+\epsilon) + \epsilon n(\log|\mathcal{X}|) + 2 \\
&= n(H+\epsilon'),
\end{aligned}$$

where $\epsilon' = \epsilon + \epsilon\log|\mathcal{X}| + \frac{2}{n}$ can be made arbitrarily small by choosing n properly.

INFORMATION THEORY & CODING